

JUNHUI XU

Ph.D. Candidate, Tsinghua University
Experimental Nuclear Physics

Ph.D. Advisor: Zhigang Xiao



Personal info

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Links

🌐 ORCID
📄 Scopus
📖 Inspire-HEP
🎓 Google Scholar

Skills

💻 C++, C, Python, Bash
⚙️ Linux (Ubuntu)
🏢 CERN ROOT, Geant4, MATLAB, SolidWorks
🔧 Detector electronics (CAEN), DAQ systems, oscilloscope, 3D printing
📊 Scientific computing, statistical analysis, and machine learning (Python, PyTorch)
✍️ LaTeX, Markdown, HTML

RESEARCH EXPERIENCE

Nucleon Short-Range Correlations

CSHINE EXPERIMENT, RIBLL-1 (HIRFL), LANZHOU OCT 2023 – PRESENT

- Participated in the self-developed Compact Spectrometer for Heavy Ion Experiment (CSHINE) at the Radioactive Ion Beam Line in Lanzhou (RIBLL-1).
- Took primary responsibility for the CsI(Tl) gamma-ray hodoscope used to detect high-energy bremsstrahlung photons from low-energy heavy-ion collisions, including energy calibration, response characterization, and spectrum reconstruction.
- Reconstructed hard photon energy spectra from the $^{124}\text{Sn}+^{124}\text{Sn}$ system at 25 MeV/u and constrained IBUU-MDI transport model simulations, extracting an SRC-induced high-momentum tail fraction of $(20 \pm 3)\%$ in ^{124}Sn nuclei.

Two-Particle Correlation Function Analysis

TSINGHUA UNIVERSITY, DEPARTMENT OF PHYSICS FEB 2024 – PRESENT

- Applied femtoscopic two-particle correlation techniques combined with the optical deconvolution algorithm to directly image emission sources and extract strong interaction parameters from hadron pairs, beyond conventional Gaussian parameterizations.
- Using pp and $\bar{p}\bar{p}$ correlation functions in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV, reconstructed proton and antiproton emission sources and constrained strong interaction parameters, demonstrating matter-antimatter symmetry and non-Gaussian source structures.
- Extended the imaging approach to three-dimensional source reconstruction and applied it to $\pi\pi$ correlation functions, laying the groundwork for extracting neutron skin properties of nuclei.

Deep-Sea Neutrino Detection

TROPICAL DEEP-SEA NEUTRINO TELESCOPE (TRIDENT) COLLABORATION JUN 2024 – PRESENT

- Performed detector response simulations for a compact deep-sea neutrino telescope array, modeling atmospheric neutrino interactions in seawater and subsequent Cherenkov photon production, propagation and event reconstruction.
- Developed a deep neural network framework based on the detector responses to classify track-like and cascade-like atmospheric neutrino events, demonstrating the feasibility of measuring atmospheric neutrino oscillation parameters in a deep-sea environment.

S π RIT Experiment

RIKEN / MICHIGAN STATE UNIVERSITY / TSINGHUA UNIVERSITY MAY 2024 – SEPT 2024

- Member of the international S π RIT collaboration at RIKEN, responsible for detector operation and electronics control during SAMURAI beam experiments.

EDUCATION

Tsinghua University

DEPARTMENT OF PHYSICS
BEIJING, CHINA AUG 2022 – JUN 2027 (EXPECTED)

Ph.D. Candidate in Physics

Research Field: Experimental Nuclear Physics
Research focus: Nucleon Short-Range Correlations (SRCs), Heavy-ion Collisions (HICs), Two-Particle Correlation Functions, Deep-sea Neutrinos

South China Normal University

SCHOOL OF INFORMATION AND OPTOELECTRONIC SCIENCE AND ENGINEERING
GUANGZHOU, CHINA AUG 2018 – JUN 2022

Bachelor of Science

GPA: 4.06/5.00 Rank: 1/137

Thesis Title: A Study of the Nuclear Medium Effect on Hadron Fragmentation Functions

Probing Nucleon Short-Range Correlations with Hard Gamma Rays in Heavy-Ion Collisions

Short-range correlations (SRCs) generate high-momentum nucleons beyond the Fermi surface in atomic nuclei. The fraction of high-momentum tail (HMT) nucleons controls the hardness of bremsstrahlung γ -ray spectra in low-energy heavy-ion collisions, with larger HMT fractions producing harder spectra. The s bremsstrahlung γ -rays, predominantly emitted from early-stage neutron-proton scattering, therefore provide a sensitive probe of SRC-induced high-momentum components.

- Junhuai Xu *et al.*, Precise measurement of short-range correlations in nuclei from bremsstrahlung gamma-ray emission in low-energy heavy-ion collisions, ***Phys. Rev. Research* 7, 043174 (2025)**.

Performed a high-precision extraction of the SRC fraction in $^{124}\text{Sn}+^{124}\text{Sn}$ collisions at 25 MeV/u by comparing measured hard bremsstrahlung γ -ray spectra with IBUU-MDI transport model calculations. Determined the SRC-induced high-momentum nucleon fraction to be $(20 \pm 3)\%$.

- Junhuai Xu *et al.*, Experimental study of bremsstrahlung gamma-ray emission and short-range correlations in $^{124}\text{Sn}+^{124}\text{Sn}$ collisions at 25 MeV/u, *Submitted to Phys. Rev. C*, under review, **arXiv:2508.04550**.

Documented the full experimental and analysis framework supporting the SRC measurement, including detector setup, calibration, trigger logic, background subtraction, and systematic uncertainty evaluation. Enabled reproducible extraction of the HMT fraction via comparison with transport model simulations.

- Junhuai Xu *et al.*, Linear response of CsI(Tl) crystal to energetic photons below 20 MeV, ***Nucl. Instrum. Meth. A* 1080, 170787 (2025)**.

Quantified the linear response of CsI(Tl) crystals to high-energy γ -rays up to 20 MeV and incorporated corrections into the CSHINE gamma ray hodoscope calibration, reducing detector-related systematic uncertainties in SRC-relevant spectra.

- Junhuai Xu *et al.*, Reconstruction of bremsstrahlung γ -ray spectrum in heavy-ion reactions with Richardson–Lucy algorithm, ***Phys. Lett. B* 857, 139009 (2024)**.

Introduced the optical deconvolution algorithm to reconstruct intrinsic bremsstrahlung γ -ray spectra without detector effects, enabling direct comparison with transport model predictions and supporting identification of SRC-induced high-momentum nucleons.

Imaging Freeze-Out Sources and Extracting Strong Interaction Parameters in Relativistic Heavy-Ion Collisions

In relativistic heavy-ion collisions, freeze-out source distributions are usually assumed Gaussian to extract source sizes and strong interaction parameters from two-particle correlations. We applied an optical deblurring algorithm to directly image proton and antiproton sources in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV without assuming a shape, simultaneously constraining strong interaction parameters. The results reveal non-Gaussian source structures and matter-antimatter symmetry at freeze-out stage in relativistic heavy-ion collisions.

- Junhuai Xu *et al.*, Imaging Freeze-Out Sources and Extracting Strong Interaction Parameters in Relativistic Heavy-Ion Collisions, ***Chin. Phys. Lett.* 42, 031401 (2025)**.

Certificates and Prizes

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| 2025 | First-class Comprehensive Scholarship, Tsinghua University |
| 2024 | Second-class Comprehensive Scholarship, Tsinghua University |
| 2024 | Outstanding Teaching Assistant, Tsinghua University |
| 2021 | Third Prize, 11th National Undergraduate Mathematics Competition, Final Round (Non-Math Major) |
| 2020 | First Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM) |

Co-authored Publications

- **Neutron skin and three-dimensional source imaging in heavy-ion collisions.** Contributed to femtoscopic studies using $\pi\pi$ correlation functions to reconstruct the three-dimensional emission source and probe neutron density distributions and neutron skin effects in heavy-ion collisions.
 - *arXiv:2510.20554* (nucl-ex).
 - ***Phys. Lett. B* 870, 139963 (2025)**.
- **CSHINE experiment.** Participated in physics analyses and detector instrumentation studies within the CSHINE experimental program.
 - ***Phys. Rev. Lett.* 134, 222301 (2025)**.
 - ***Nucl. Instrum. Meth. A* 1070, 170055 (2025)**.
- **S π RIT experiment.** Contributed to detector operation and electronics development for the S π RIT at RIKEN.
 - ***Nucl. Instrum. Meth. A* 1083, 171158 (2026)**.

 Google Scholar (click to view full publication list)